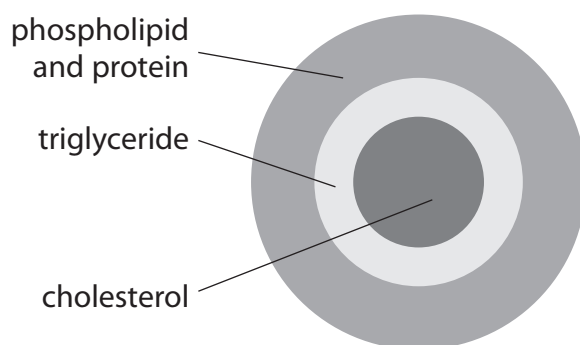


8 Low-density lipoproteins (LDLs) transport lipids around the body in the blood.

Low-density lipoproteins can result in the development of atherosclerosis.

They can be absorbed into the endothelial cells lining arteries and broken down by free radicals.

The diagram shows a low-density lipoprotein containing cholesterol.



(a) Compare and contrast the structure of a triglyceride and a phospholipid.

(3)

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- (b) Explain why the properties of LDLs enable cholesterol to be transported in the blood.

(3)

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- (c) The diameters of LDLs range from 19 nm to 24 nm.

The table shows some information about LDLs.

Diameter of LDL / nm	Volume of LDL / nm ³	Volume of cholesterol / nm ³	Ratio of LDL volume to cholesterol volume
19	3590	523	7:1
24		523	

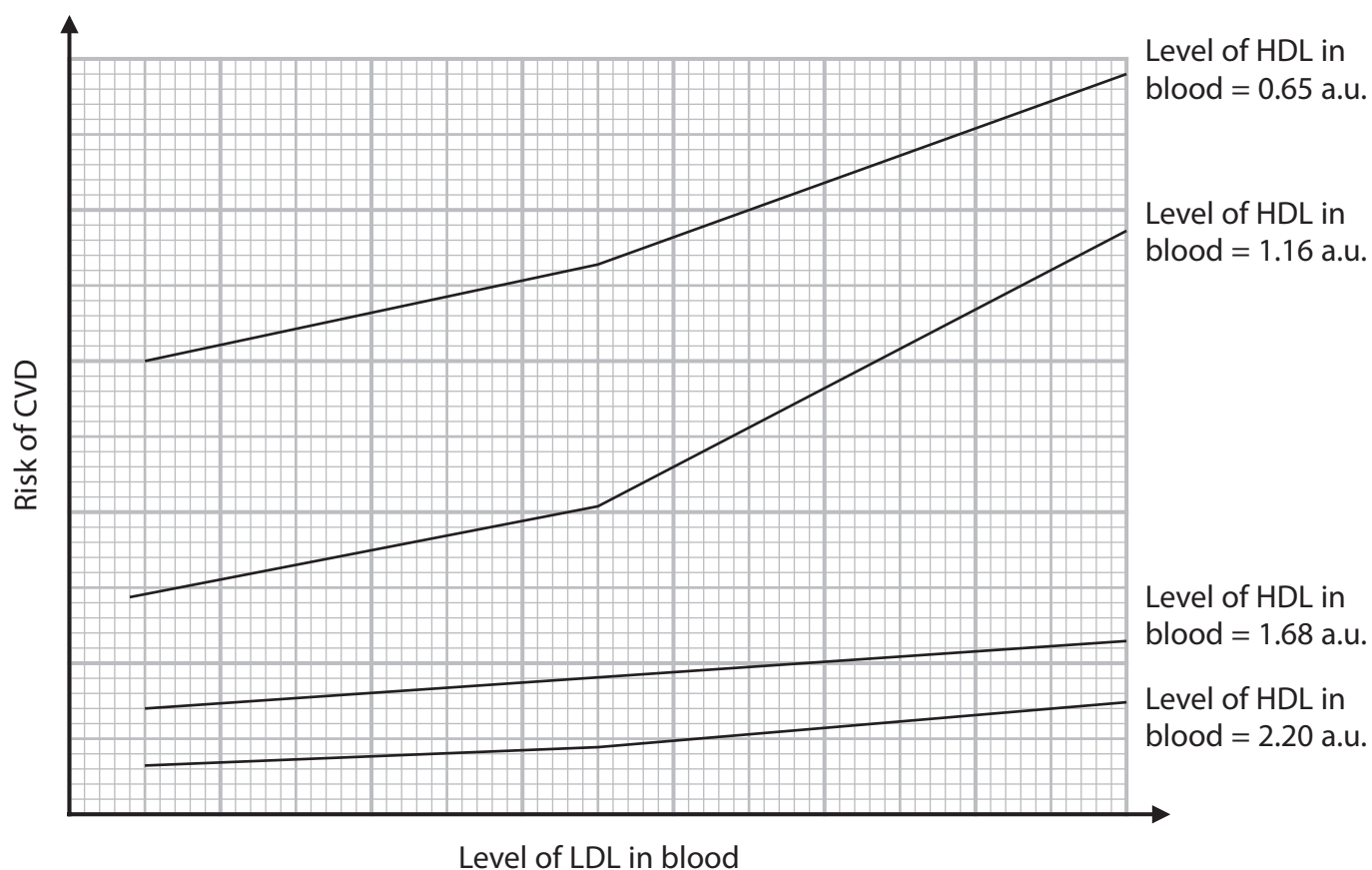
- (i) Complete the table by calculating the volume of LDL and the ratio of LDL volume to cholesterol volume.

Use the formula $v = \frac{4}{3}\pi r^3$

(3)



*(ii) The graph shows the relationship between LDLs, high-density lipoproteins (HDLs) and the risk of CVD.



Explain why measuring only the level of LDL in the blood is **not** a reliable predictor of CVD.

Use the graph, all the information in this question and your own knowledge to support your answer.

(6)

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(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS



P 6 2 4 5 6 A 0 2 9 3 2